

Module 1: [Based on class & objects]

Problem Statement: Write a C++ program using class and objects to automate the banking process with following constraints:

- i. A new account can be created and an old account can be deleted
- ii. An account holder can debit/credit money
- iii. An account can't be deleted if money exists in the account
- iv. The program should be following menu operated
- v. The private data members of an account holder's class are "Account No", "Name", "Age" and "Balance".
- vi. Design member methods for individual operation
- vii. Produce appropriate message when required

a. The template of the class is

```
class Account{  
    //private data members  
    //public methods  
};
```

b. The menu is :

```
***** Main Menu *****  
  
    1. Open New Account  
    2. Close Old Account  
    3. Deposit Money  
    4. Withdraw Money  
    5. Check Balance  
    6. Exit  
    Enter Your option:(1-6):_
```

c. Few Snapshots:

1. Deposit Money

```
Enter your option(1-6):3 <enter>  
Enter Account Number:12678 <enter>  
Enter Amount: 5000 <enter>  
Successfully deposited  
Press any key go to main menu.....
```

Code:

```
#include <iostream>
using namespace std;
class Account
{
public:
    string name;
    int acn;
    int balance;

    Account()
    {
    }

    void Open_account(string n, int a, int b)
    {
        name = n;
        acn = a;
        balance = b;
    }

    void Delete()
    {
        if (balance > 0)
            cout << "Account can not be deleted. Withdraw your money first ;)" << endl;
        else
        {
            name = " ";
            acn = 0;
            balance = 0;
            cout << "Account deleted successfully" << endl;
        }
    }

    void deposit(int b)
    {
        int old = balance;
        balance = b;
        old += balance;
        balance = old;
    }

    void withdraw(int b)
    {
        int old = balance;
        balance = b;
```

```

        if (b > old)
            cout << "Insufficient balance " << endl;
        else
        {
            old -= b;
            balance = old;
        }
    }
    void Getdata()
    {
        cout << "Account number : " << acn << endl;
        cout << "Name          : " << name << endl;
        cout << "balance        : " << balance << " Taka " << endl;
        cout << ".....\n";
    }
};

void print()
{
    cout << "***** Main Menu ***** \n";
    cout << "1. Open New Account " << endl;
    cout << "2. Close Old Account" << endl;
    cout << "3. Deposit money" << endl;
    cout << "4. Withdraw money" << endl;
    cout << "5. Check balance" << endl;
    cout << "6. Exit" << endl;
    cout << "   Enter your option (1-6): " << endl;
}

int main()
{
    int choice;
    int index = 0;
    string Name;
    int Acn;
    int Bal;
    int temp_Acn, temp_balance;

    Account b[10000];

    while (1)
    {
        print();
        cin >> choice;

        switch (choice)
        {
            case 1:
                cout << "Enter your Name: ";
                cin >> Name;

```

```

    cout << "Enter your balance: ";
    cin >> Bal;
    Acn = 1000 + index + 1;

    b[Acn].Open_account(Name, Acn, Bal);
    b[Acn].Getdata();
    index++;
    break;
case 3:
    cout << "Enter your account number: ";
    cin >> temp_Acn;
    if (temp_Acn < (index - 1) && temp_Acn > 1000)
    {
        cout << "Enter your deposite balance: ";
        cin >> temp_balance;
        b[temp_Acn].deposit(temp_balance);
        b[temp_Acn].Getdata();
        cout << "Success" << endl;
    }
    else
        cout << "Invalid account" << endl;

    break;
case 4:
    cout << "Enter your account number: ";
    cin >> temp_Acn;
    if (temp_Acn < (index - 1) && temp_Acn > 1000)
    {
        cout << "Enter your withdrwal balance: ";
        cin >> temp_balance;
        b[temp_Acn].withdraw(temp_balance);
        b[temp_Acn].Getdata();
        cout << "Sucess" << endl;
    }
    else
        cout << "Invalid account\n";

    break;
case 2:
    cout << "Enter your account number: ";
    cin >> temp_Acn;
    if (temp_Acn < (index - 1) && temp_Acn > 1000)
    {
        b[temp_Acn].Delete();
    }
    else
        cout << "Invalid account" << endl;

    break;

```

```

    case 5:
        cout << "Enter your account number: ";
        cin >> temp_Acn;
        if (temp_Acn < (index - 1) && temp_Acn > 1000)
        {
            cout << "Your current balance: " << b[temp_Acn].balance << " Taka" <<
endl;
        }
        else
            cout << "Invalid account" << endl;

        break;
    case 6:
        cout << "Thank you" << endl;
        return 0;
        break;

    default:
        break;
}
}
}

```

Output Snippet

```

1. Open New Account
2. Close Old Account
3. Deposit money
4. Withdraw money
5. Check balance
6. Exit
    Enter your option (1-6):

```

```

1
Enter your Name: Ragib
Enter your balance: 5000
Account number : 1001
Name           : Ragib
balance        : 5000 Taka
.....

```

Enter your option (1-6): 3
Enter your account number: 1001
Enter your deposit balance: 6000
Account number : 1001
Name : Ragib
Balance : 11000 Taka
.....
Success

Enter your option (1-6): 4
Enter your account number: 1001
Enter your withdrawal balance: 3000
Account number : 1001
Name : Ragib
Balance : 8000 Taka
.....

Enter your option (1-6): 2
Enter your account number: 1001
Account can not be deleted. Withdraw your money first ;)

Enter your option (1-6): 5
Enter your account number: 1001
Your current balance: 8000 Taka